

Nathan Burns

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EDUCATION

University of Cincinnati

Expected Graduation: May 2026

B.Sc., Software Application Development | Masters: Business Administration | Minor: Computer Science

- GPA: 3.989
- Relevant Coursework: Data Structures & Algorithms, Computer Programming II, Machine Learning, AI Applications, Web Server Development, Mobile Device Programming, Enterprise Development, Database Mgmt I and II, Programming Languages
- Teaching Assistant for Data Technologies Programming and Database Management II

SKILLS

Programming Languages: C#, Python, Java, C++, TypeScript, JavaScript, Go, HTML/CSS, SQL

Frameworks: Angular, React, Next.js, Express.js, FastAPI, .NET Core, ASP.NET, Xamarin, RxJS, Bootstrap, Tailwind CSS, Entity, Jest

Software: Git, Docker, Postman, NGINX, AWS EC2, Vercel, MongoDB, SQLite, SSMS, Jira, Azure DevOps, PostgreSQL, Supabase

EXPERIENCE

London Computer Systems

Software Developer Intern

May 2023-August 2023 | May 2024-August 2024 | January 2025-August 2025

- Contributed to the development of our flagship product, *Rent Manager Express*, a web app for property management.
- Led frontend development utilizing the **Angular** framework, executed backend functionality using **C#** and **ASP.NET**, and implemented efficient database management through **MySQL** and **HeidiSQL**. Successfully completed 160+ tickets.
- Collaborated in an **Agile** environment, actively participating in daily stand-up meetings, sprint planning, and iterative development to deliver high-quality features and enhancements on time.
- Led the enhancement of Accounts Receivable Task Automation by adding a scheduling option that sends notifications on specific days, introducing a filter for the newest unpaid charges, and improving the user interface for a more intuitive setup.
- Refined the Surveys feature by updating the kebab menu with options to view, void, delete, and copy surveys. Designed and implemented a new Send Survey tool for targeted tenant feedback, streamlining the process.
- Developed a 'CAM Expense Adjustments' report with versatile display options for tenant-based CAM cost allocation.

QA Software Tester Intern

May 2022-August 2022

- Developed and executed detailed test cases on software updates to our company's product.
- Collaborated with developer teams to complete 80+ tests, contributing to the successful implementation of software updates.

PROJECTS

CLI and Web Productivity Tracker

August 2025

- Developed a macOS **CLI tool** in **Go** using **AppleScript** to track active apps, combined with a custom **Chrome extension** using **Javascript** and **Chrome's Tabs API** to monitor website usage. All activity logs were stored and managed in **PostgreSQL**.
- Implemented a background daemon using **launchd** for reliable scheduled data collection, building CLI commands with **Cobra** and designing terminal-based reports with **text/tabwriter** and **asciigraph** to visualize detailed usage metrics.
- Built a local **React** dashboard with **Next.js** and **Recharts** to visualize tracked data, hosted with backend services using **Docker**.

ActiveLife Tracker

June-August 2024

- Designed and developed a full-stack [fitness tracking web app](#) using the **MEAN** stack (**MongoDB**, **Express.js**, **Angular**, **Node.js**)
- Deployed the frontend via **Vercel** and hosted the backend on **AWS EC2**, configuring **NGINX** as a reverse proxy with SSL certificates from **Certbot** to enable secure HTTPS communication and custom domain routing with **pm2** for robust uptime.
- Engineered a secure custom auth system using **JWTs** and **bcrypt**, with server-side validation, email verification, and tokenized password reset flows via **BREVO**. Integrated adaptive **Google reCAPTCHA** for bot prevention and login abuse mitigation.
- Implemented **caching** mechanisms on the frontend to minimize redundant API calls, improving app responsiveness and overall performance. Designed **algorithms** that generate personalized macro plans dynamically based on user profile data.
- Built interactive, dynamic charts for daily, monthly, and yearly exercise and nutritional breakdowns using **Chart.js** and **Tailwind CSS** and integrated the **FDA FoodData Central API** to provide real-time food item lookup and nutritional data.
- Managed backend **API rate limiting** with **express-rate-limit** middleware to protect services from abuse and ensure reliability.
- Applied reactive state management using **RxJS** to handle asynchronous data flows and user interactions smoothly.
- Conducted API and backend testing using **Postman** and **Jest**, including unit and integration tests with in-memory MongoDB.

Automated Sports Text Script

July 2023

- Developed a **Python** script to interact with the **ESPN API** for retrieving daily sports data and rigorously tested with **Postman**.
- Integrated **Twilio API** to send automated sports texts, deploying the script daily on a **Raspberry Pi** with **Crontab** as a server.